

Physiology of the digestive system

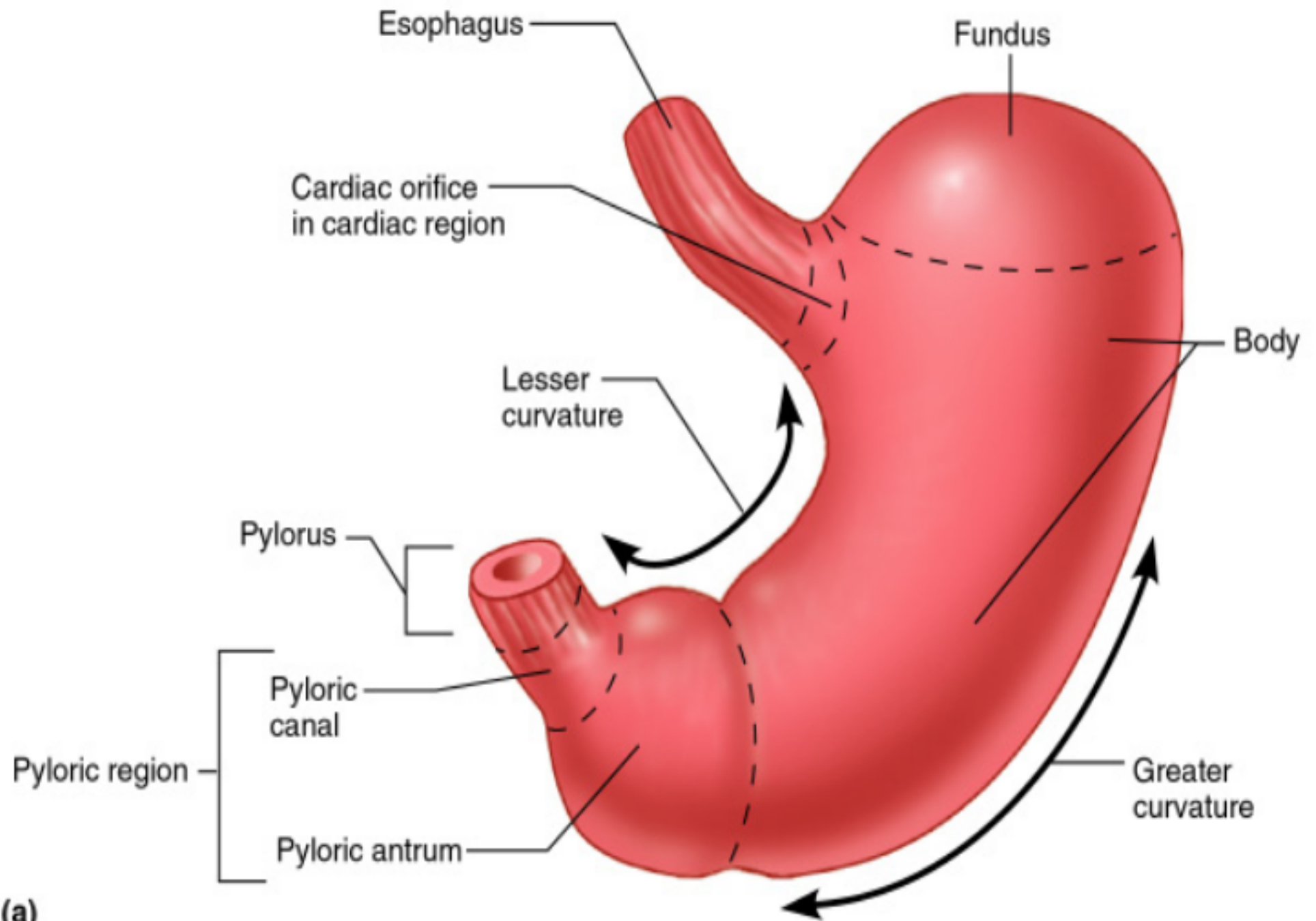


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Functional anatomy

- **3d** organ in the tract
- **3** layers
- **3** types of muscles
- Food stays in it for about **3** hours
- Controlled by **3** systems (sympathetic, parasympathetic and hormonal)
- **3** types of gastric glands
- Gastric juice amount per day is **3** liters
- pH of the gastric juice is around **3**



Gastric glands and their secretions

- Mucous cells (goblet): secretes mucous.
- Parietal cells: secretes HCL & intrinsic factor.
- Chief cells: secretes pepsin enzyme for protein digestion.
- Mnemonic: Chief of Pepsi. p for pH



Gastric pit



Cell Types	Substance Secreted
Goblet cells	Mucus (protects stomach lining)
Parietal cells	Gastric acid (e.g. hydrochloric acid)
Chief cells	Pepsinogen (protease precursor)
D cells	Somatostatin (inhibits acid secretion)
G cells	Gastrin (stimulates acid secretion)



Other secreting cells in the stomach

- G-cells : Release **Gastrin**
- Enterochromaffin-like cells (ECL), secrete **Histamine**
- D cell : Release **Somatostatin**

**THESE ARE HORMONES AND NOT FOUND
IN THE GASTRIC JUICE**



Functions of the stomach

- Food storage
- Mechanical and chemical digestion start in the stomach. HCL creates and acidic environment which is unfavorable to microorganisms.
- HCL also maintains iron in the ferrous state to facilitate its absorption.



Functions of the stomach

- Production of intrinsic factor (IF) to facilitate absorption of vitamin B12.
- Hormonal function: produces the hormone Gastrin to increase HCL production.



Functions of HCL

- Breaks connective tissues and muscle fibers, which helps with the digestion
- Activates pepsinogens (optimum pH)
- Provide an optimum pH for action of pepsin
- Bactericidal properties
- Helps in absorption of iron and calcium



Gastric juice

- *Volume:* 2–3 L/day.
- *pH:* Highly acidic. Around 3
- *Osmolarity:* Isotonic (same as the blood)

Contents:

- Water & ions (Protons, chlorine,,,,,) (Inorganic)
- Intrinsic factor (organic)
- Digestive enzymes (pepsin) (organic)

Remember: NO HORMONES



Pepsinogens

- Inactive precursors of pepsin
- Activated by HCL
- Optimum PH 1.8–3.5
- Pepsins hydrolyses protein into short chain peptides



Gastric lipase

- This enzyme hydrolyses triacylglycerol
- It acts mainly on triacylglycerol with short chain fatty acids
- It is much less important than pancreatic lipase



Intrinsic factor

- It is secreted by parietal cells
- It is glycoprotein that combine with vitamin B 12 protect it through its passage in the small intestine
- It helps in absorption of the vitamin in distal part of ileum



Mucous

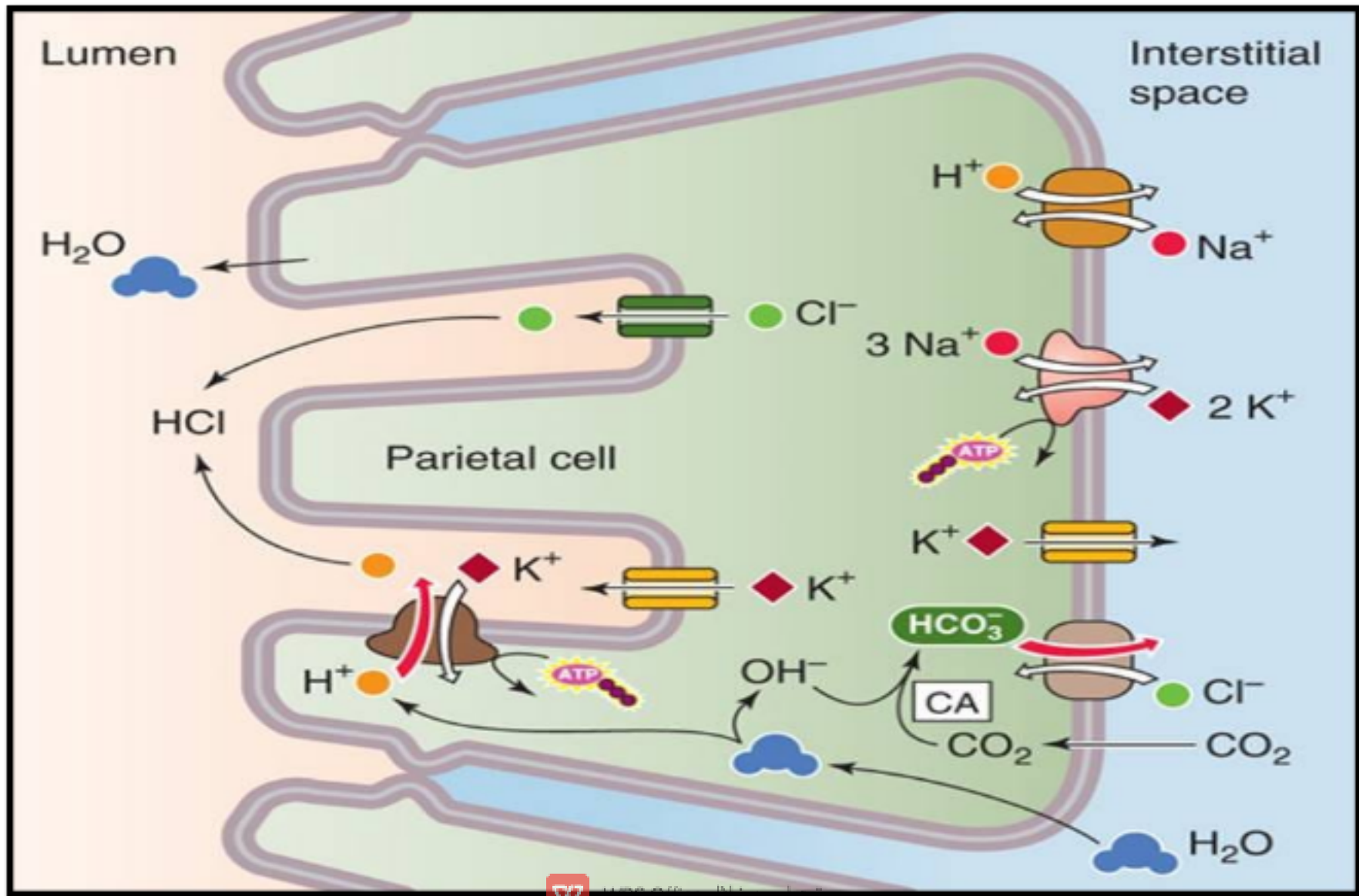
- Secreted by mucous neck cells
- its main function is protection of stomach wall against acid and enzymes
- Other factors in protection:
 - Bicarbonate
 - Prostaglandins
 - Phospholipid layer of cell membrane



The making and secreting of the HCL

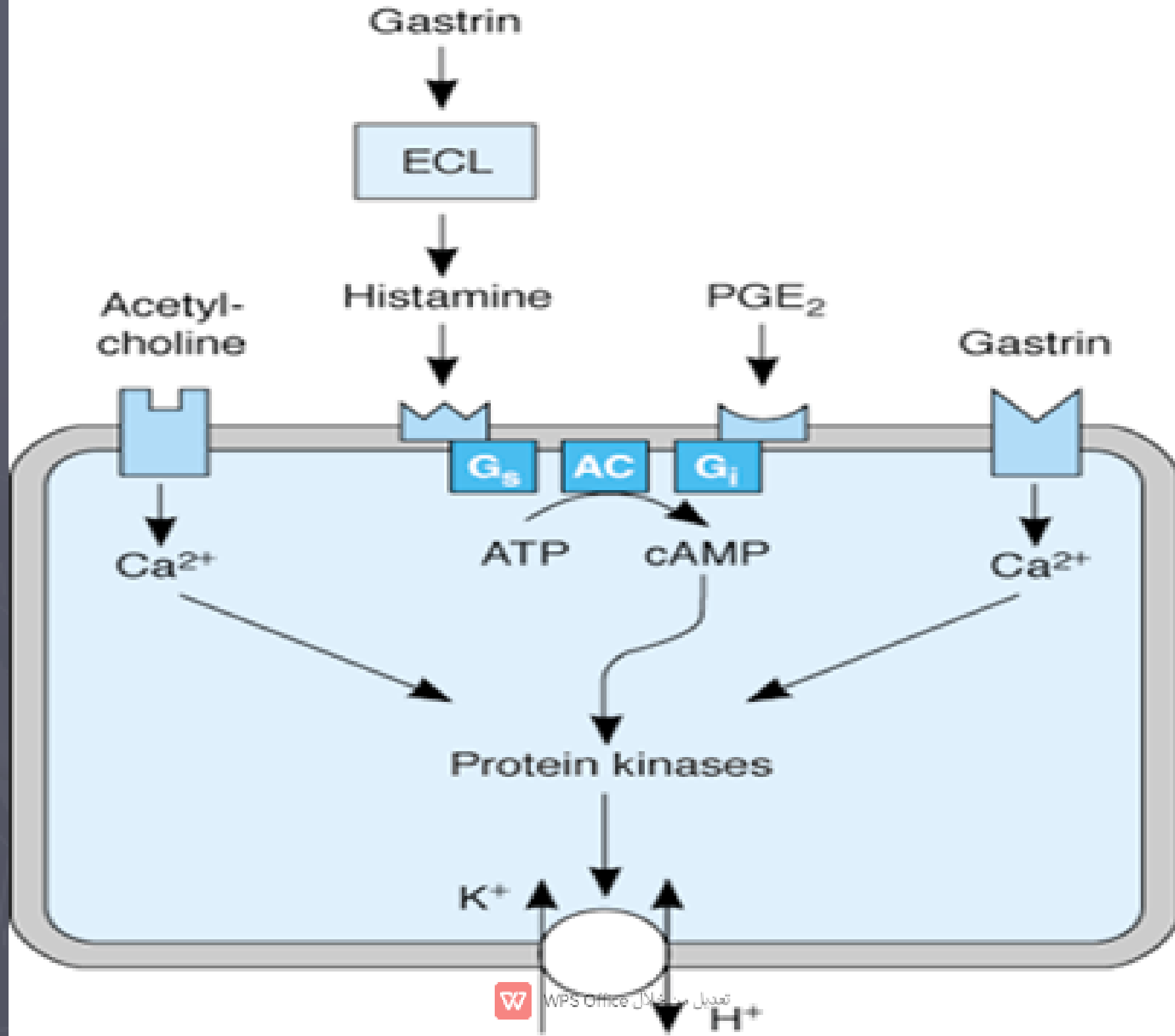


Physiology of the parietal cell



Stimulation of HCL secretion





Hormones ↓ HCL secretion

- Somatostatin
- Secretin
- Glucagon
- VIP
- GIP
- Enterogastone
- Prostaglandin



Control of gastric secretion

- Control is either hormonal or neural (ANS)
 - Neural mechanism
 - Hormonal mechanism
- Which are activated through 3 phases
 1. cephalic phase (20%)
 2. gastric phase (70%)
 3. intestinal phase(10%)



Cephalic phase

- Initiated by thought, sight, taste or smell of food
- Secretes 20% of gastric juice
- The aim is to prepare stomach for food
- The vagus nerve stimulates Ach release



Gastric phase

➤ Initiated by entry of the food into the stomach & cause distention which stimulate secretion :

1. Neuraly:

-Vago vagal reflex, integrated in the brain stem

2. hormonally:

- gastrin that stimulate secretion release in response to :

- gastric distention

- amount of aa & peptides directly stimulate parietal cells (chemoreceptors in the mucosa)



Intestinal phase

- Gastric secretion initially inhibited when food entering the duodenum
- The acidity in the small intestine detected by pH receptors
- H^+ , FAA & $\uparrow$ osmolarity stimulate release of enterogastrones from the duodenum
- GLP : inhibit both HCL & gastrin release



Other influences

- Hypoglycemia stimulates acid and pepsin secretion (via the vagus)
- Alcohol (Directly)
- Caffeine (Directly)



See you next lecture

